Roberto (André) Mossi Milla

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SUMMARY

Software Engineer experienced in artificial intelligence, full-stack development, computer vision, distributed systems, and real-time simulation environments. Published researcher with experience spanning academic research, industrial computer vision systems, AI-powered healthcare platforms, multimodal AI workflows, entrepreneurship, and cross-platform software architecture.

EDUCATION

Gannon University, Pennsylvania, United States

Bachelor of Science in Computer Systems Engineering, May 2025[11]

ISI Japanese Language School, Tokyo, Japan

Intensive Japanese Language Program, October 2025 - March 2026[9]

Harvard University, Online

CS50's Introduction to Computer Science, December 2022[12]

Extreme Networks, Online

Introduction to Future Networks, May 2022[1]

RESEARCH EXPERIENCE

Gannon University, Pennsylvania, United States

Research Assistant, August 2024 - May 2025

- Conducted research under faculty supervision on multi-agent artificial intelligence simulating natural selection and survival behaviors in dynamic environments.[10]
- Implemented NeuroEvolution of Augmenting Topologies (NEAT) to evolve neural network architectures and behavioral strategies across generations.
- Independently implemented NeuroEvolution of Augmenting Topologies (NEAT) and its genome representation in C++, and developed a 3D multi-agent simulation environment in Unreal Engine with procedural world generation and physics-based interactions.
- Used CUDA to support GPU-accelerated simulation and experimentation within the evolutionary computation research workflow.
- Research paper accepted and published by the American Society for Engineering Education (ASEE) and presented at a Sigma Xi conference at Penn State University.[6] [3]

INDUSTRY EXPERIENCE

Ingeniería Digital, Tegucigalpa, Honduras

Full-Stack Developer / AI Systems Engineer, April 2026 - Present

AI-Assisted Medical Document Generation Platform

- Built an AI-assisted medical document generation platform for post-surgery reports (IDR), reducing report preparation from approximately one week to 1 day and final document generation to under one minute. Combined OCR extraction, LLM-assisted grammar correction, and professional document structuring using Next.js, React, TypeScript, PostgreSQL, AWS S3, Amazon Textract, and OpenAI APIs.
- Architected authenticated document management workflows using Auth0, enabling users to generate, analyze, edit, version, and manage AI-generated reports through a full-stack web platform.
- Designed scalable file-processing pipelines integrating OCR extraction, cloud storage, AWS S3 buckets, and PostgreSQL persistence for intelligent document retrieval and processing.
- Implemented internationalization workflows and automated translation asset generation using CMake to support multilingual deployments across US and Honduran clients.

AI Vehicle Damage Assessment Platform

- Developed an AI-assisted vehicle damage assessment platform that reduced assessment time from more than one week to approximately 1–2 minutes, combining VIN-history analysis, image-based damage detection, and automated insurance estimation workflows.

- Integrated AWS Rekognition to localize vehicles in uploaded images and Gemini Flash to detect and analyze damage, with Claude APIs and OpenAI APIs supporting language-model workflows across document and vehicle-assessment processes.
- Developed data acquisition services using Playwright to collect and process vehicle information required for valuation and insurance analysis workflows.
- Implemented interactive damage annotation and review tools using Konva, allowing users and insurance personnel to identify, visualize, and validate detected vehicle damage regions.
- Developed responsive client-facing and administrative dashboards using React, Next.js, TypeScript, PostgreSQL, AWS S3, and Shadcn/UI.
- Maintained engineering standards and CI/CD workflows using GitHub Actions, Husky, ESLint, Prettier, lint-staged, and Commitizen.

Gannon University, Pennsylvania, United States

Computer Vision Engineer (Project-Based) Industry Partner (NDA), August 2024 - May 2025

- Architected and deployed a real-time industrial traceability system using Ultralytics YOLOv8 to detect and track pallets, automatically generating structured events with images and detection metadata.[4]
- Collaborated in a cross-functional team contributing across hardware, backend, and software components; took primary ownership of the monitoring dashboard and visualization interface.
- Developed a full-stack web dashboard integrated with backend services and TimescaleDB to visualize detection events, images, and operational analytics in real time.
- Implemented interactive annotation and administrative tools to correct bounding boxes, curate datasets, and support iterative model retraining and image-processing workflows.
- Containerized distributed services using Docker and integrated multi-language components (C++, Python, Rust) to coordinate inference, data processing, and communication.

Dinant, Tegucigalpa, Honduras

Full-Stack Developer, June - July 2023

- Led the end-to-end design and development of two production web platforms, conducting requirements gathering, architecture design, client presentations, and deployment.[2]
- Architected a contract lifecycle management platform supporting device sales, rentals, repairs, and multi-country operations.
- Developed a multi-role web application using Oracle APEX, JavaScript, HTML, and CSS supporting digital signatures, PDF generation, and administrative reporting.
- Built a factory maintenance management platform enabling machine monitoring, maintenance workflows, repair tracking, inventory reporting, and procurement insights.

PROJECTS & ENTREPRENEURSHIP

IRIS, Founder, Product Designer & Software Engineer — 2026 - Present

- Founded and lead development of IRIS, an AI-powered computer vision platform designed to provide accessible vision intelligence through a scalable API and user-facing software platform.
- Architected a cloud-based computer vision infrastructure supporting authenticated API access, image processing workflows, model inference, and scalable interaction between client applications and AI services.
- Designed multimodal AI workflows combining computer vision models and large language models to transform visual inputs into structured, actionable information.
- Developed the platform architecture around secure API authentication, user-specific access tokens, asynchronous processing, and scalable backend services for integrating computer vision capabilities into external applications.
- Designed the product as both a developer-facing API platform and consumer-facing application, enabling third-party software to integrate IRIS computer vision capabilities while providing direct access to AI-powered workflows.
- Established engineering workflows for reproducible model deployment, API development, containerized infrastructure, and continuous integration and delivery.

NOIR, Founder, Product Designer & Software Engineer — 2025 - Present

- Founded and lead development of NOIR, a cinema-focused software platform designed as a continuity layer across streaming platforms, physical theaters, and consumer devices, unifying movie discovery, availability, playback, and cross-device viewing experiences.
- Designed the product architecture around a platform-agnostic coordination model, integrating movie metadata, streaming availability, authentication, user profiles, session state, and device continuity without owning or redistributing third-party content.
- Authored a comprehensive technical strategy and product design document defining the system architecture, phased product roadmap, monetization strategy, platform integration model, compliance requirements, and international expansion strategy.
- Architected a distributed backend platform using Rust, SeaORM, Axum, PostgreSQL, Redis, Supabase, Docker, and Railway to support authentication, real-time synchronization, movie metadata aggregation, availability services, and persistent user state.
- Developed native SwiftUI applications targeting macOS, iOS, iPadOS, and watchOS with real-time synchronization, push notifications, subscription management, cross-device session continuity, and AI-assisted natural-language movie discovery.
- Designed an AI intelligence layer that translates natural-language viewing intent into structured movie discovery and availability queries, connecting recommendations directly to actionable viewing paths rather than isolated search results.
- Designed engineering workflows utilizing GitHub Actions, Linear, Commitizen, SwiftLint, Clippy, Docker, and automated CI/CD pipelines.

Copilot-Lualine, Open Source Project

- Designed and developed Copilot-Lualine, an open-source Neovim plugin integrating GitHub Copilot status information into the Lualine statusline ecosystem. It holds as of now 117 stars and going up.[5]
- Implemented Lua-based editor tooling focused on workflow visibility, developer productivity, and Neovim customization.

LEADERSHIP & ORGANIZATIONS

- Founder, Product Designer & Software Engineer, NOIR (2025 – Present).
- Founder, Product Designer & Software Engineer, IRIS (2025 – Present).
- Member, Mathematics Club, Gannon University (2023 - 2025).
- Member, Computer Club, Gannon University (2022 - 2025).
- President, Computer Club, Dowal School (High School).
- Member, Leo Club, Dowal School (High School).

PUBLICATIONS

- Mossi, R. A. — "Tag AI-Sandbox," Proceedings of the ASEE Annual Conference, 2025 (peer-reviewed).[6]

TECHNICAL SKILLS

- Languages: C++, Rust, Swift, TypeScript, JavaScript, Python, Java, SQL, Lua, Bash, PowerShell.
- Frontend: SwiftUI, React, Next.js, Vite, Tailwind CSS, Shadcn/UI.
- Backend & Cloud: Axum, SeaORM, Tokio, PostgreSQL, Redis, Supabase, AWS S3, Amazon Textract, Docker, Auth0, Railway, REST APIs.
- AI & Computer Vision: OpenAI APIs, Gemini Flash, Claude APIs, AWS Rekognition, Amazon Textract, Ultralytics YOLOv8, CUDA, NEAT, OCR Pipelines, Multimodal AI Workflows.
- Developer Tooling: GitHub Actions, Husky, ESLint, Prettier, Commitizen, CMake, SwiftLint, Clippy, Git.
- Operating Systems: Linux, macOS, Windows.
- Languages: Spanish, English, Japanese.

AWARDS AND HONORS

- "School Development Award" Award obtained for outstanding development of a videogame using Blueprints in Unreal Engine 4, 2020[8]
- "School Development Award" Award obtained for outstanding development of a videogame using C++ and Unreal Engine 4, 2016[7]

REFERENCES

[1] Extreme Academy. *Extreme Networks Certification*. Dec. 1, 2022. URL: https://andremossi.vercel.app/experiences/extreme (visited on 02/15/2026).

[2] Rigoberto Funes. *Dinant Recommendation Letter*. Feb. 11, 2026. URL: https://andremossi.vercel.app/PDF/Recommendations/dinant.pdf (visited on 02/15/2026).

[3] lah585. *Tag AI-Sandbox Presentation*. Sigma Xi. Mar. 11, 2025. URL: https://sites.psu.edu/behrendsigmaxi2025/2025/03/11/tag-ai-sandbox/ (visited on 02/15/2026).

[4] Andre Mossi. *Computer Vision Project*. Feb. 15, 2026. URL: https://andremossi.vercel.app/experiences/traceability (visited on 02/15/2026).

[5] Andre Mossi. *Copilot Lualine*. Nov. 28, 2023. URL: https://github.com/AndreM222/copilot-lualine (visited on 08/26/2026).

[6] Roberto Andre Mossi and Ramakrishnan Sundaram. "Tag AI-Sandbox Paper". In: *2025 ASEE North Central Section (NCS) Annual Conference*. 10.18260/1-2–54692. Marshall University, Huntington, West Virginia: ASEE Conferences, Mar. 2025. URL: https://peer.asee.org/54692.

[7] Dowal School. *Software Engineer 2016 Award*. May 12, 2016. URL: https://andremossi.vercel.app/experiences/dowal2016 (visited on 02/15/2026).

[8] Dowal School. *Software Engineer 2020 Award*. May 12, 2020. URL: https://andremossi.vercel.app/experiences/dowal2020 (visited on 02/15/2026).

[9] ISI Japanese Language School. *ISI Japanese School*. Mar. 10, 2026. URL: https://andremossi.vercel.app/experiences/isi (visited on 06/02/2026).

[10] Ram Sundaram. *Tag AI-Sandbox Recommendation Letter*. Feb. 20, 2025. URL: https://andremossi.vercel.app/PDF/Recommendations/ai-sandbox.pdf (visited on 02/15/2026).

[11] Gannon University. *Gannon University BS Diploma*. May 10, 2025. URL: https://andremossi.vercel.app/experiences/gannon (visited on 02/15/2026).

[12] Harvard University. *Harvard CS50 Certification*. Dec. 20, 2022. URL: https://andremossi.vercel.app/experiences/harvard (visited on 02/15/2026).